ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 1.1.1 C ✓ 1.1.2 A ✓ 1.1.3 B ✓ 1.1.4 D ✓ 1.1.5 A ✓ 1.1.6 C ✓ 1.1.7 C ✓ 1.1.8 A ✓ 1.1.9 D ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 A ✓ 1.3.2 F ✓ 1.3.3 G ✓ 1.3.4 I ✓ 1.3.5 B ✓ 1.3.6 C ✓ 1.3.7 H ✓ 1.3.8 J ✓ 1.3.9 D ✓ 1.3.10 E ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 (a) Cartesian – three linear joints ✓, rectangular box-shaped envelope ✓ (b) cylindrical – one rotary base joint and two linear joints ✓, cylinder-shaped envelope ✓ (c) articulated – rotary joints (usually six) ✓, roughly spherical / irregular envelope ✓ (6)

2.2 (a) SCARA or delta robot ✓ – very fast, accurate horizontal pick-and-place ✓ (b) articulated (6-axis) robot ✓ – can reach around and into the car body at any angle ✓ (4)

2.3 Any THREE named ✓ and explained ✓: world (base) frame – fixed to the robot base / cell, used for general positions; joint coordinates – each axis moved individually by angle; tool frame – based on the tool centre point, moves with the tool, used to approach parts along the tool direction; user (work-object) frame – defined on a fixture or table, so a program can be moved by redefining the frame (6)

2.4 Accuracy – how close the robot gets to the commanded target position ✓✓; repeatability – how close it returns to the same position each time it repeats the movement ✓✓ (a robot can be repeatable without being accurate). (4)

[20]

QUESTION 3

3.1.1 $x = 30\cos 30° + 20\cos 90° \checkmark = 25{,}98 + 0 \checkmark = 25{,}98\ \text{cm} \checkmark$; $y = 30\sin 30° + 20\sin 90° \checkmark = 15 + 20 \checkmark = 35\ \text{cm} \checkmark$ (6)

3.1.2 $30 + 20 = 50\ \text{cm} \checkmark$ (1)

3.2 There may be more than one set of joint angles (or none) for a position ✓, and the equations are non-linear and harder to solve ✓. (2)

3.3 Point-to-point: only the start and end positions matter; the path between is not controlled ✓, e.g. spot welding, pick and place ✓; continuous path: the tool follows a controlled path all the way ✓, e.g. arc welding, painting, gluing ✓ (4)

[13]

QUESTION 4

4.1 $V_{avg} = 0{,}75 \times 12$ ✓ $= 9$ V ✓ (2)

4.2 (a) $90° \div 1{,}8°$ ✓ $= 50$ steps ✓ (b) $1\,000 \div 200 = 5$ revolutions ✓; 5×2 mm ✓ $= 10$ mm ✓ (5)

4.3 DC motor: speed control by voltage/PWM, position needs extra feedback ✓ – wheels of a mobile robot ✓; servo motor: built-in feedback, moves to a set angle ✓ – grippers, steering, small arm joints ✓; stepper motor: moves in exact steps without feedback (open loop) ✓ – 3D printers, CNC axes, Cartesian robots ✓ (6)

4.4 A slotted / marked disc on the shaft turns between a light source and a sensor ✓; each slot produces a pulse ✓; the controller counts the pulses (and their sequence) to calculate the angle, speed and direction ✓. (3)

4.5 It sends a sound pulse and measures the time until the echo returns; the sound travels to the object and back ✓. $d = (v \times t) \div 2$ ✓ $= (343 \times 0{,}0029) \div 2$ ✓ $= 0{,}50$ m $(49{,}74$ cm$)$ ✓ (4)

[20]

QUESTION 5

5.1 Forward: both wheels turn forward at the same speed ✓; turn left: the right wheel turns faster than the left wheel ✓; spin on the spot: wheels turn at the same speed in opposite directions ✓ (3)

5.2 Any THREE (method ✓ + advantage ✓): teach pendant – simple and intuitive, positions taught on the real robot; offline programming / simulation – no production downtime, test safely; lead-through (hand guiding) – fast for complex paths such as painting, no coding needed; text-based programming (e.g. RAPID, Python) – precise, logic and calculations possible (6)

5.3 Any FOUR with purpose ✓✓✓✓: LiDAR – map the corridor and avoid obstacles; ultrasonic sensors – detect nearby objects / people; bump sensors – stop on contact; wheel encoders – odometry; camera – recognise doors, signs, people; IMU – orientation; cliff sensors – detect stairs (4)

5.4 Any FOUR with a task ✓✓✓✓: two-finger gripper – picking parts; vacuum gripper – lifting boxes or glass; magnetic gripper – steel sheets; welding torch – welding; spray gun – painting; screwdriver – assembly (4)

[17]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE